ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ROBOTICS FUNDAMENTALS

PAPER 2 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Paper 2 (design related). Section A: objective questions (30) · Section B: 3D printing, basic electricity and the multimeter, microcontrollers and single-board computers, breadboard prototyping and sensors, GPIO programming with Python, and a robot design task (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
5. A non-programmable calculator may be used.
6. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The 3D printing method that pushes melted plastic filament through a hot nozzle, layer by layer, is …

A stereolithography (SLA). B selective laser sintering (SLS).
C laser cutting. D fused deposition modelling (FDM).

1.1.2 Building an object by adding material layer by layer is called …

A subtractive manufacturing. B additive manufacturing.
C casting. D milling.

1.1.3 The software that cuts a 3D model into layers and creates the printer instructions is a …

A compiler. B browser. C slicer. D spreadsheet.

1.1.4 The file type most often exported from CAD software for 3D printing is …

A .stl B .docx C .mp3 D .py

1.1.5 The instrument used to measure voltage, current and resistance is a …

A thermometer. B multimeter. C oscilloscope only. D tachometer.

1.1.6 A Raspberry Pi is best described as a …

A microcontroller board. B sensor.
C motor driver. D single-board computer (microprocessor).

1.1.7 The pins on a Raspberry Pi that can be programmed as inputs or outputs are called …

A GPIO pins. B HATs.
C USB ports. D jumper wires.

1.1.8 A sensor that detects the movement of warm bodies such as people is a …

A force-sensing resistor. B photoresistor.
C PIR sensor. D thermistor.

1.1.9 In conventional current flow, current flows from …

A negative to positive. B ground to the supply.
C positive to negative. D the load to the battery only.

1.1.10 An add-on board that fits on top of a Raspberry Pi's GPIO header is called a …

A HAT. B shield only. C breakout. D relay.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 PLA filament is made from plant-based material such as corn starch.

1.2.2 ABS usually needs a lower printing temperature than PLA.

1.2.3 Support structures hold up overhanging parts of a model while it prints.

1.2.4 A closed circuit has a complete path for current to flow.

1.2.5 To measure current, a multimeter is connected in parallel with the component.

1.2.6 An Arduino Uno runs a full desktop operating system.

1.2.7 The rows of holes in each half of a breadboard are connected to each other inside the board.

1.2.8 LEDs must be connected the right way round because they are polarised.

1.2.9 Python 3 is the version of Python prescribed for this subject.

1.2.10 An analogue-to-digital converter changes a digital signal into a voltage.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Plastic wire that feeds an FDM printer	A	physical computing
1.3.2	Flat surface that the first layer is printed on	B	filament
1.3.3	Removing supports and sanding a printed part	C	breadboard
1.3.4	Component that limits the current in a circuit	D	jumper wire
1.3.5	Board for building circuits without soldering	E	buzzer
1.3.6	Short wire with pins used to connect components	F	Ohm's law
1.3.7	Relationship V = I × R	G	resistor
1.3.8	Small board that makes one sensor easy to connect	H	print bed
1.3.9	Output device that makes a sound	I	breakout
1.3.10	Programming real hardware such as LEDs and buttons	J	finishing

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: 3D PRINTING

2.1 List, in the correct order, the FOUR main stages of the 3D printing process. (4)

2.2 A learner wants to print a small cover for a robot's battery box. Compare PLA and ABS under the headings printing temperature, strength / heat resistance and ease of printing, and recommend one. (5)

2.3 Name FOUR main components of an FDM printer and give the function of each. (4)

2.4 Give TWO advantages and TWO disadvantages of 3D printing parts for robots. (4)

2.5 Explain why the first layer must stick well to the print bed, and name ONE way to improve bed adhesion. (3)

[20]

QUESTION 3: ELECTRICITY AND THE MULTIMETER

3.1 Use the water-flow analogy to explain voltage, current and resistance. (3)

3.2 A red LED needs 2 V and 20 mA. It is powered from a 5 V supply through a series resistor.

3.2.1 Calculate the voltage across the resistor. (1)

3.2.2 Calculate the value of the resistor. (3)

3.2.3 Calculate the power used by the resistor. (2)

3.3 Explain how a multimeter must be connected to measure (a) the voltage across the LED and (b) the current through it. (2)

3.4 A learner's circuit does not work. Explain how the multimeter's continuity test can help to find an open circuit. (2)

[13]

QUESTION 4: COMPONENTS, BREADBOARDS AND SENSORS

4.1 Compare an Arduino Uno and a Raspberry Pi under the headings type of board, operating system and typical robot use. (6)

4.2 Explain how the holes of a breadboard are connected, including the power rails. (3)

4.3 Two identical LEDs (each with its own resistor) are connected to a battery, first in series and then in parallel. Explain what happens to the other LED in EACH case if one LED fails (breaks open). (4)

4.4 Name the input sensor you would use for EACH situation: (a) switching on a light when someone walks into a room; (b) detecting when a robot is standing on a pressure pad; (c) measuring how bright it is. (3)

4.5 Explain why a Raspberry Pi needs an analogue-to-digital converter (ADC) to read a temperature sensor that gives an analogue voltage. (2)

4.6 Differentiate between a passive and an active electronic component, and give ONE example of each. (4)

[22]

QUESTION 5: PROGRAMMING AND DESIGN

5.1 Study the Python program below, written for a Raspberry Pi using the gpiozero library, and answer the questions.

```
from gpiozero import LED, Button
from time import sleep

led = LED(17)
button = Button(2)

while True:
    if button.is_pressed:
        led.on()
        sleep(3)
        led.off()
    sleep(0.1)
```

5.1.1 To which GPIO pins are the LED and the button connected? (2)

5.1.2 Describe what the program does when the button is pressed. (2)

5.1.3 Explain the purpose of the line while True: (1)

5.1.4 Rewrite the lines inside the if-statement so that the LED flashes on and off FIVE times (0,5 s on and 0,5 s off). (4)

5.2 You must design a simple "robot doorbell": when a visitor presses a button, a buzzer sounds for 2 seconds and an LED stays on for 10 seconds. List the hardware components you need and draw a flowchart of the program. (6)

[15]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER